



**WHITEPAPER: ESTABLISHING STANDARDS FOR RESOURCE-
EFFICIENT AI SOFTWARE & COMPUTE MANAGEMENT**

CHENYI ANG, CHAIRMAN

Chartered Software Developer Association (CSDA)

07 FEBRUARY 2025

Executive Summary

Artificial Intelligence (AI) is at the forefront of global technological transformation, driving advancements in automation, decision-making, and large-scale data processing. However, these innovations come at a cost—AI development and deployment require significant computational power, leading to high energy consumption, carbon emissions, and environmental impact.

The emergence of a compute marketplace, where computing power is traded as a commodity, has raised concerns about unregulated energy usage, inefficient compute distribution, and sustainability. The recent article in *The Wall Street Journal* titled "*AI Needs a Lot of Computing Power—Is a Market for Compute the Next Big Thing?*" highlights the growing demand for computational resources and the risks associated with excessive consumption, cost inflation, and lack of accountability.¹

This whitepaper outlines the need for an international standard to govern resource-efficient AI software and compute usage. The proposed ISO standard aims to establish benchmarks for energy-efficient AI workloads, standardize carbon measurement methods, and encourage sustainable AI development practices.

Table of Contents

1. The Compute Market & Rising AI Energy Demands	3
1.1 Growth of Compute-as-a-Commodity	3
1.2 Environmental & Economic Challenges	3
2. The Role of a Global Standard for Resource-Efficient AI Software	3
2.1 Setting Efficiency Benchmarks	3
2.2 Standardizing Energy & Carbon Measurement	4
2.3 Optimizing Software-Hardware Interaction	4
2.4 Supporting Policymakers & Regulators	4
3. Addressing the Compute Market: Regulation vs. Innovation	4
4. The Path Forward: Implementing a Resource-Efficient AI Standard	4
4.1 Collaboration with Standards Bodies	4
4.2 Industry & Government Adoption	4
4.3 Next Steps	5
5. Conclusion & Call to Action	5
Next Steps & How to Get Involved	5

¹ The Wall Street Journal (2025) 'AI needs a lot of computing power: Is a market for compute the next big thing?', *The Wall Street Journal*, 7 February. Available at: <https://www.wsj.com/tech/ai/ai-needs-a-lot-of-computing-power-is-a-market-for-compute-the-next-big-thing-2302133c>

WHITEPAPER: ESTABLISHING STANDARDS FOR RESOURCE-EFFICIENT AI SOFTWARE & COMPUTE MANAGEMENT

1. The Compute Market & Rising AI Energy Demands

1.1 Growth of Compute-as-a-Commodity

With AI models requiring massive computational resources, companies are turning to cloud providers, specialized AI chips (GPUs/TPUs), and decentralized compute markets. While this offers accessibility, it also introduces:

- Unregulated energy consumption, leading to higher carbon emissions.
- Market inefficiencies, where demand outstrips supply, inflating costs.
- Lack of transparency, as AI developers have no standardized benchmarks for energy-efficient computing.

1.2 Environmental & Economic Challenges

The AI industry's compute demand is unsustainable without regulation. Key concerns include:

- **Excessive carbon footprint:** Training an advanced AI model can consume as much energy as five cars over their lifetime.²
- **Cooling & heat waste:** High-performance GPUs generate significant heat, increasing cooling costs for data centers.
- **Unfair access:** Smaller AI companies struggle to afford expensive compute resources, widening the AI divide.

2. The Role of a Global Standard for Resource-Efficient AI Software

A resource-efficient AI standard is crucial for guiding AI developers, cloud providers, and policymakers in sustainable compute usage. The proposed standard aims to:

2.1 Setting Efficiency Benchmarks

- Define power usage metrics for AI model training (e.g., kWh per epoch).
- Establish guidelines for GPU/TPU energy efficiency, reducing unnecessary computation.

² Strubell, E., Ganesh, A. & McCallum, A. (2019) *Training a single AI model can emit as much carbon as five cars in their lifetimes*. MIT Technology Review. Available at: <https://www.technologyreview.com/2019/06/06/239031/training-a-single-ai-model-can-emit-as-much-carbon-as-five-cars-in-their-lifetimes/>

2.2 Standardizing Energy & Carbon Measurement

- Create universal reporting methods for AI compute emissions.
- Introduce Carbon Usage Effectiveness (CUE) & Power Usage Effectiveness (PUE) standards.

2.3 Optimizing Software-Hardware Interaction

- Encourage low-energy AI model designs (e.g., model pruning, quantization).
- Promote resource-aware cloud AI deployment to minimize redundant compute cycles.

2.4 Supporting Policymakers & Regulators

- Provide a framework for sustainable AI governance in compute-intensive applications.
 - Help governments align AI policies with United Nations Sustainable Development Goals (SDGs).
-

3. Addressing the Compute Market: Regulation vs. Innovation

While the compute market provides flexibility for AI companies, it lacks transparency and sustainability mechanisms. The proposed standard would help:

1. **Prevent overconsumption:** Companies would optimize compute allocation to reduce unnecessary AI workloads.
 2. **Create fairer access:** Standardized efficiency benchmarks help smaller AI firms compete without excessive compute costs.
 3. **Encourage sustainable AI investments:** Investors & cloud providers favor green-compliant AI models, reducing carbon emissions.
-

4. The Path Forward: Implementing a Resource-Efficient AI Standard

4.1 Collaboration with Standards Bodies

- Work with the International Organization for Standardization (ISO) on ISO/IEC JTC 1/SC 39 (Sustainability for IT) & ISO/IEC JTC 1/SC 42 (AI Standards) to formalize energy-efficient AI software metrics.
- Engage AI cloud providers (AWS, Google Cloud, Microsoft Azure) to implement best practices in sustainable AI deployment.

4.2 Industry & Government Adoption

- Encourage data centers to comply with AI sustainability benchmarks.
- Advocate for AI compute efficiency policies similar to energy efficiency standards in hardware.

4.3 Next Steps

- Develop an open-source benchmarking tool for AI compute efficiency.
 - Publish a technical guideline for AI engineers on energy-efficient coding.
 - Seek collaboration with AI industry leaders, cloud providers, and policymakers.
-

5. Conclusion & Call to Action

AI computing demands will continue to grow, but without standardized governance, the environmental impact will escalate unsustainably. A global standard for resource-efficient AI software is critical to:

- Ensure energy-efficient AI development.
- Regulate the compute marketplace to avoid resource overconsumption.
- Encourage transparent reporting of AI compute emissions.

Policymakers, AI researchers, and industry leaders must act now to establish standards that balance innovation with sustainability. By adopting resource-efficient AI software practices, we can drive responsible AI development while minimizing its environmental footprint.

White Paper published 07 February 2025

Next Steps & How to Get Involved



For collaboration inquiries or contributions, contact:
Chenyi Ang

Chairman of Chartered Software Developer Association (CSDA)

Email: chenyi@charteredsoftwaredeveloper.org

About Chartered Software Developer Association (CSDA)

Chartered Software Developer Association is a global professional association for technology profession. CSDA brings together leading standards technology practitioners & researchers for industry-wide cooperation on setting standards needed for AI & ESG revolution by advancing responsible AI and mapping out the coding strategies that can reduce CO2 emissions.

<https://charteredsoftwaredeveloper.org/>